Nurun Nahar Fiha

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Education

Jahangirnagar University **Savar, Bangladesh**
M.Sc. in Computer Science and Engineering *Jun 2024 – Sep 2025*

- **CGPA: 3.83/4.00 (Rank: 7/44)**
- **Notable Courses:** Natural Language Processing, Machine Intelligence & Expert Systems, Network Security, Distributed Database, Pattern Recognition

B.Sc. in Computer Science and Engineering *Feb 2019 – May 2024*

- **CGPA: 3.69/4.00 (Rank: 9/59)**
- **Notable Courses:** Machine Learning, AI, Data Structures & Algorithms, Database Systems, Operating Systems, HCI

Research Experience

Multilingual Sentiment Analysis of Social Media Reactions to OpenAI's Ghibli-Style Image Generation Feature

[Manuscript in preparation] — M.Sc. thesis; supervised by Dr. Israt Jahan

- Studied how the public perceives AI-generated art, conducting one of the first large-scale sentiment analyses of a viral generative-AI feature, by building a 36.7K-record multilingual corpus of Twitter and YouTube reactions with translation, emoji-aware preprocessing, and VADER-based weak supervision.
- Benchmarked 21 transformer and hybrid Transformer–RNN configurations to identify the best accuracy-efficiency trade-off for monitoring fast-moving social discourse; BERT-base performed best (91.27%), with a lightweight DistilBERT–BiGRU hybrid matching it at a fraction of the compute.

Deep Learning Based Bangladeshi Currency Coin Recognition

[Published at ICREST 2025, IEEE Xplore] — B.Sc. thesis; supervised by Md. Golam Moazzam

- Addressed coin identification for visually impaired users, as Bangladeshi coins are nearly identical in size and texture, by building a 2,800-image, 14-class dataset captured under real-world lighting and rotation conditions.
- Trained a Faster R-CNN (ResNet-50 FPN) against a CNN baseline in PyTorch, reaching 91.18% test accuracy (vs. 84.04%).

A Comparative Analysis of Low-Resource Techniques for Named Entity Recognition in Bengali

M.Sc. course research project (NLP)

- Investigated the effectiveness of modeling approaches for Bengali, a widely spoken but low-resource language, by benchmarking CRF, BiLSTM-CRF, mBERT, and XLM-RoBERTa on a custom 25K-sentence dataset merged from three public corpora.
- Enhanced models with FastText embeddings, character-level CNNs, and linguistic features; XLM-RoBERTa achieved the best F1 (0.758).

Cyberbullying Detection on Social Media Platforms Using Machine Learning

B.Sc. course research project (ML)

- Relabeled a 16K-comment Bengali toxic-comment dataset into 3 classes (Non-Bullying, Bullying, Crime) and compared four classifiers (LR, DT, RF, XGBoost). XGBoost with class-imbalance weighting performed best, with 75.4% test accuracy and a 0.761 10-fold CV score.

Awards & Recognition

National Science and Technology (NST) Fellowship | Ministry of Science and Technology **2025 – 2026**

- Awarded a $500 NST Fellowship under the Physical Science category to support M.Sc. research.

2nd Runner-up, Zelf Hackathon 2.0 **Oct 2024**

- Placed 3rd among 300+ candidates in a QA-focused hackathon, leading to a recruitment interview and subsequent hire.

Merit Scholarship | Jahangirnagar University **2019 – 2025**

- Received $100/year merit scholarships during both B.Sc. and M.Sc. for academic excellence (Top 10%).

Competitive Programming **2019 – 2024**

- Solved 700+ problems on LeetCode, Codeforces, and UVa; participated in national and international programming contests including ICPC 2021, ADA Lovelace NGPC 2021, and NGPC 2021 (ICT Division).

Test Scores & Certifications

IELTS Academic: Overall 7.5 (Listening: 8.0, Reading: 8.5, Writing: 6.5, Speaking: 6.0) **Jun 2026**

Fundamental Information Technology Engineer (FE) Examination – Full Passer **Oct 2023**

- Ranked **35th** nationally in the ITEE, a national-level IT engineers' examination governed by IPA, Japan.

Work Experience

Software Quality Assurance Engineer | Zelf AI | Remote Brooklyn, NY, USA | **Dec 2024 – Present**

- Perform root-cause analysis of bugs via Google Cloud logs, database queries, and API testing; systematically audited the platform, reporting 50+ previously undetected defects.
- Build and maintain automated test suites in Playwright alongside manual functional, regression, and UAT cycles; track issues in Jira, coordinating with developers and stakeholders.

Junior SQA Engineer | Dynamic Solution Innovators Dhaka, Bangladesh | **Apr 2024 – Dec 2024**

Project: KeyPleaz – a US-based app connecting real estate agents instantly.

- Served as the sole QA engineer on the project – authored and managed 500+ test cases in Qase, covering functional, regression, smoke, sanity, and UAT cycles.
- Owned end-to-end testing of high-risk features including in-app chat, calling, and Stripe payment transactions across multiple devices, identifying edge-case failures before release.
- Presented features and demos directly to the US client and coordinated with PM, BA, and developers on user story reviews, ensuring requirements aligned with implementation.

Projects

OBE-Based Result Management System | *Python, Django, Bootstrap, SQLite* [GitHub]

- Built a role-based academic results platform where students view results, teachers analyze courses via auto-generated CO-PO attainment graphs, and administrators monitor performance department-wide.

Konta Niben – an E-commerce System | *MERN, Tailwind CSS, Jest, GitHub Actions* [GitHub]

- Developed a full-stack e-commerce platform following Agile/SDLC practices across 2 sprints, with JSDoc documentation, Jest unit tests, and CI via GitHub Actions.

JU CSE Routinery | *Java, Android SDK, SQLite* [GitHub]

- Built an Android app for organizing CSE department class routines for individual batches.

BoiPoka | *React, Tailwind CSS* [GitHub]

- Developed a front-end bookstore application for browsing and discovering books.

Skills & Technologies

Languages: Python, C++, C, Java, JavaScript, SQL
ML/AI: PyTorch, scikit-learn, OpenCV, NLTK, Hugging Face Transformers, NumPy, Pandas, Matplotlib
Web Development: React, Django, Node.js, REST APIs, HTML5, CSS3, Bootstrap 5, Tailwind CSS
Testing & Automation: Playwright, Cypress, Postman, Qase
Databases: PostgreSQL, MySQL, MongoDB
Tools & Platforms: Git & GitHub, Linux, LaTeX, VS Code, Azure Boards, Jira/Trello, Slack

Extracurricular & Leadership Activities

Founding Member | JU Robotics Club **Feb 2023 – Oct 2024**

- Co-founded the club; built a 6-DoF robotic arm and line follower robots, and competed in inter-university events including Technocrats V.1 (IUBAT).

Publication Secretary | JU Computer Club **Jun 2021 – Apr 2024**

- Promoted from Executive Member to Publication Secretary within a year; managed the club's publication team for nearly 3 years, designing posters and promotional materials for events and recruitment drives.